Day 11 · Week 3 · Investigate and synthesise

Finding and judging evidence

Where is the evidence on our problem, and how far can we trust it?



Day 11 of 20

Today

By the end of today, you will have

Entered and rated at least six sources in your evidence register.

- Navigate the main kinds of Indian source, from surveys and administrative data to audits.
- Judge a source by who produced it, how, when, at what level and with what limits.
- Distinguish what a source establishes from what it only suggests.
- Cite sources accurately in Chicago style, and never invent a detail.

Competencies: E2 · E3 · J5

Session plan

10 min	**Opening problem**
20 min	**Concept: the evidence ecosystem**
15 min	**Demonstration: rating sources**
30 min	**Activity: the evidence hunt**
35 min	**Studio: own evidence register**
10 min	**Debrief and exit task**

Opening problem

- **Three sources say different things about your problem: a government dashboard, a CAG audit and a newspaper investigation. Which do you believe?**

Before you answer, write what you would need to know about each.

The Indian evidence ecosystem

Type	Examples	Good for	Watch for
Household surveys	National Family Health Survey; Periodic Labour Force Survey; consumption surveys	Representative estimates of conditions and behaviour	Often valid at state level only; rounds years apart
Administrative data	UDISE+; scheme MIS dashboards; municipal records	Coverage, inputs and outputs, often by district	Records what the system reports, not what happened
Outcome surveys	ASER (Pratham); National Achievement Survey	Outcomes rather than inputs	Different methods give different pictures
Audit and oversight	CAG reports; social audits	Gaps between rules and practice	Specific places and periods; may be dated
Parliamentary	Standing committee reports; PRS analyses	Budget scrutiny and official responses	Government answers are claims, not findings
Research and journalism	Academic studies; reputable reporting	Mechanisms, voices, context	Case selection; check method and sources

Five questions to judge any source

1. **Who produced it, and why?**
   A department reporting on its own scheme has reasons to look good.

2. **How was it produced?**
   A sample survey, an administrative count, an audit test or a case study.

3. **When, and for which period?**
   Check the reference year, not just the publication year.

4. **At what level is it valid?**
   Many surveys are representative only at state level.

5. **What does it leave out?**
   Who is missing: informal workers, migrants, unregistered households?

Rating sources for Madhyapur

Source	Type	Reliability	Reason
Solid Waste Management Rules, 2016	Law	High	The primary text of the rules
City waste management plan (illustrative)	Official plan	Medium	States intentions, not performance
Ward compliance returns (illustrative)	Administrative	Low to medium	Self-reported by staff who are judged on them
A CAG performance audit of waste management in the state	Audit	High for its period	Tests records against practice; may be several years old
Case study of a city working with a waste pickers' cooperative	Research	Medium	Rich on mechanism; one city, may not transfer
Newspaper report of crews mixing waste	Journalism	Low to medium	Direct observation, but one ward on one day

Reliability is a judgement with a reason, not a label.

What a source establishes, and what it only suggests

Only suggests

- ○ That households know the rules
- ○ That the same is true in Madhyapur today
- ○ That the share is the same in your district

Establishes

- ● The rules require separation at source
- ● The audit found records did not match practice in the cities it sampled
- ● The survey estimates a share for the state, with a margin of error

Write each claim at the strength its evidence supports. The Ledger tag follows from this.

Citing in Chicago style

Kind	Format
Government report	Ministry. Title of Report. New Delhi: Government of India, year.
Audit report	Comptroller and Auditor General of India. Report No. N of year: Title. New Delhi: CAG, year.
Journal article	Surname, First name. "Title." Journal volume, no. issue (year): pages.
Dataset	Organisation. Dataset name, round (year). Accessed date. URL.

Never invent a page number, report number or URL. If you cannot verify it, leave it out and flag it.

Activity: The evidence hunt

Find and rank credible sources fast, and learn which kinds of source answer which kinds of question.

Scenario

Each pod gets one question and twenty minutes. How has the share of children in a chosen state who can read a Grade 2 text changed over the last decade? Are sanctioned posts in primary health centres in a chosen state filled? What has the CAG said about solid waste management in a chosen state? How much did a chosen state budget for school meals last year, and how much did it spend?

What to do

1. Find five sources on your question in twenty minutes.
2. For each, record type, producer, period and level.
3. Rate each source's reliability, with a reason.
4. Choose the single most useful source and say why.
5. Report in two minutes; your evidence checker defends one rating.

Time 30 minutes

Groups Pods of four, one question per pod

Output Five rated sources and the most useful one, justified

Your case: Worksheet 5.1

Building the evidence register

1. Add at least six sources to your register, covering at least three types.
2. For each, record producer, method, period, level and limitations.
3. Rate reliability, with a reason.
4. Note which hypothesis each source supports or weakens.
5. Mark whether you checked the original and whether AI helped.

Hand in

Updated evidence register by the start of Day 12.

Quality check

At least six sources.

Tag every claim in your Analyst's Ledger: fact, inference, assumption or hypothesis.

What to take away

1. Each kind of source is good for some questions and poor for others.
2. Judge a source by who made it, how, when, at what level and what it leaves out.
3. Write each claim at the strength its evidence supports.
4. Cite the original, verify every detail and never invent one.

Exit task Name your most reliable source and its biggest limitation.

Before the next session

Independent work

Read your two most important sources properly, including the methods section. Write three sentences on what each establishes and what it does not. About 50 minutes.

Read

The methods chapter or note of your main survey or administrative source.

Next

Day 12: What the numbers say

Tomorrow: what the numbers actually say. Bring your data.